

Molecules

Assignment #2

1 Electron eigenvalues of the oxygen molecule O₂

Calculate the electronic energy levels of the O₂ molecule as a function of the distance d between the two O ions. Initially set ISPIN = 1, non-spin polarized calculation.

Discuss the trend of the eigenvalues and total energy of the system. Based on your results construct the molecular orbitals of O₂ identifying bonding and antibonding orbitals and their occupancy.

- « Use a large cell (e.g. $10 \times 10 \times 10$) containing one atom in the origin (0,0,0) and vary the position of the second atom ($d,0,0$), with d ranging from 0.75 Å to 4.0 Å, using steps of $\Delta d = 0.25$ Å.
- « Use the GGA functional
- « Use a bash script (will be provided) to range over all values of d and perform the various calculations automatically.

2 Ground state structure and binding energy of molecules

2.1. Manual relaxation. Calculate the ground state structures of O₂ and CO.

Is the spin polarization (ISPIN=2) required or not? Hint: count the number of electrons and distribute them on the available orbitals.

Compare the results for the DFT ground-state bond-length d^{DFT} with the experimental values d^{exp} in Table 1. Report the absolute values and the relative deviations in a table.

Extract the binding energy of the O₂ molecule, by comparing the ground state energy and the energy calculated for the dissociated molecule (from Exercise 1). The total free energy can be found in the OUTCAR file ('free energy' or 'TOTEN')

Table 1: Experimental data on bond lengths d^{exp} and vibration frequencies ν .

	H ₂	O ₂	N ₂	CO	NO
d^{exp} (Å)	0.741	1.201	1.098	1.128	1.151
ν (THz)	131.9	47.4	70.7	65.1	59.3

- « By using a bash script, vary the interatomic distance d in a small range (*e.g.*, between $\pm 10\%$ and $\pm 30\%$) around the corresponding experimental d^{exp} value.
- » In order to determine the DFT ground-state bond-length d^{DFT} , fit the total energies using a polynomial function of the second order; consider a small range around the expected d^{DFT} value.

2.2. (optional) Automatic relaxation by VASP. Perform an automatic structure optimization for all the H_2 , O_2 , N_2 , CO and NO molecules, in order to determine the ground-state bond-length d_{auto}^{DFT} internally in VASP (with a single execution of VASP per molecule), by enabling the ionic relaxation mode.

Discuss the differences regarding the ground state properties (*i.e.*, the bond length) between the manual and fully-automatic approaches for the molecules considered in Exercise 2.1.

Analyze the dependence of the result for the bond-length d_{auto}^{DFT} as a function of the different functional (LDA and GGA).

- « Construct a POSCAR file by using a reasonable value for the interatomic distance (*e.g.*, the previously determined d^{DFT} value).
- « Enable the ionic relaxation (IBRION=2), with a sufficient number of steps (*e.g.*, NSW=40) in the INCAR file.
- « The choice of POTIM (scaling constant for the ionic relaxation steps) can be critical for molecules. You can test values obtained with the default value (0.5) with those obtained for POTIM=0.01.
- « Extract from the CONTCAR file the bond-length d_{auto}^{DFT} obtained in the automatic relaxation calculation.

(Optional): Why is it suggested to start from a "reasonable value for the interatomic distance" in your opinion?

3 Molecular vibrational frequency

Calculate the vibrational frequencies of O_2 and CO in the harmonic approximation.

Compare your vibrational frequencies calculated from the quadratic fit of the energy values with the experimental values in Table 1, and report the deviations.

- » Use the energy values from Exercise 2 to calculate the vibrational frequency ν of molecules in the harmonic approximation:

$$V(d) = V(d_0) + \frac{1}{2} \ddot{V}(d_0)(d - d_0)^2, \quad (1)$$

where $V(d)$ is the energy given by the interatomic distance d , and d_0 is the ground-state bond-length (*e.g.*, d^{DFT}). Obviously, $\dot{V}(d_0) = 0$. Note that this approximation holds only for small perturbation $d - d_0$ around the ground state.

- » The relationship between vibrational frequency ν and angular frequency ω of the

harmonic oscillator is given by:

$$\nu = \frac{\omega}{2\pi}, \text{ with } \omega = \sqrt{\frac{k}{\mu}}. \quad (2)$$

The force constant k of the bond is defined as $k = \ddot{V}(d_0)$, *i.e.*, the pre-factor of the quadratic term; μ denotes the reduced mass of the molecule.

3.1. (Optional) Anharmonic effects. Calculate the oscillation frequencies of O₂ and CO considering anharmonic effects.

- » Consider the Morse potential V_M (rather than the harmonic approximation) to fit the energy values, defined as

$$V_M(d) = c(1 - e^{-a(d-d_0)})^2 + b, \quad (3)$$

where a , b and c are factors which need to be identified. As starting value for the fit, select d_0 and b as obtained from the harmonic approximation.

- » The relationship between vibrational frequency ν and angular frequency ω is given by:

$$\nu = \frac{\omega}{2\pi}, \text{ with } \omega = a\sqrt{\frac{2c}{\mu}}. \quad (4)$$

Calculate the transition energy from the quantum mechanical ground-state E_0 to the first excited state E_1 .

How effective are the anharmonic effects? Does the anharmonic contribution affect equally the O₂ and CO molecules?

- » The eigenvalues E_n , solutions of the Morse potential are given by:

$$E_n = \hbar\omega \left(n + \frac{1}{2}\right) - \frac{\hbar^2\omega^2}{4c} \left(n + \frac{1}{2}\right)^2 \quad (5)$$

4 (optional) Bond-order model

Calculate the bond order b of the Li₂, Be₂, B₂, C₂, N₂, O₂, and F₂ molecules.

- « Perform fully automatic relaxations to determine the molecular ground-state bond-length (include IBRION=2, NSW=20 and POTIM = 0.1 in the INCAR file and start from a bondlength of 1.2 angstrom for all molecules, eventually a larger for the lighter atoms). If the molecule is not expected to be spin-polarized it is recommended to adopt a non-spin-polarize calculation (ISPIN=1) in order to avoid to be trapped in spurious magnetic solution (the ground state will always be associated with the lowest energy).

» Discuss the trend with a simple bond-order model:

$$b = \frac{1}{2} (N_B - N_A) , \quad (6)$$

where b is the bond order, N_B and N_A are the number of electrons in the bonding and antibonding states.

Comment on optional exercises: Please feel free to work on this only if you have the time and interest. You can still achieve the maximum grade even without completing the optional parts.